

- 2 A roller coaster car of mass 380 kg slides down a smooth slope and executes a loop of diameter d . After the loop, it experiences friction only in the area denoted as zone A, as shown in Fig. 2.1.

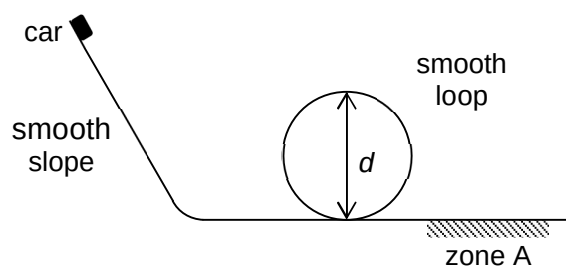


Fig. 2.1

The variations of gravitational potential energy and kinetic energy of the car with time t are shown in Fig. 2.2.

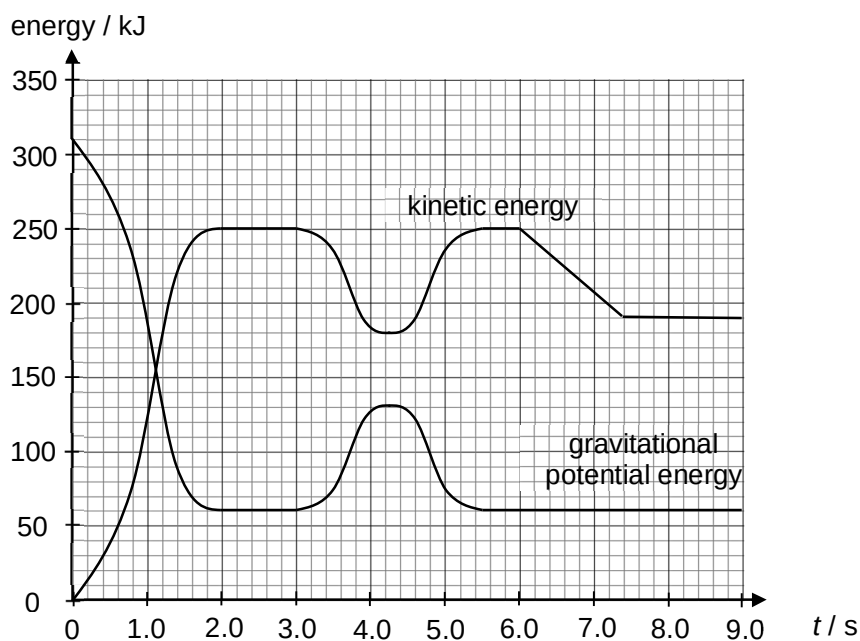


Fig. 2.2

- (a) On Fig. 2.2, sketch the variation of total mechanical energy with time of the car, from $t = 0$ s to $t = 9.0$ s. [2]

(b) Between $t = 3.0$ s and $t = 5.4$ s, the car executes the loop.

Using Fig. 2.2, calculate d .

$d = \dots\dots\dots$ m [2]

(c) (i) State the start and end times at which the car is within zone A.

start time = $\dots\dots\dots$ s

end time = $\dots\dots\dots$ s [1]

(ii) Calculate the rate at which the car loses kinetic energy in zone A.

rate = $\dots\dots\dots$ W [2]

(iii) Calculate the speed of the car when it first enters zone A.

speed = $\dots\dots\dots$ m s⁻¹ [2]

(iv) Hence, calculate the magnitude of the frictional force acting on the car when it first enters zone A.

force = $\dots\dots\dots$ N [2]